Number Theory and Cryptography

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Division Theory

Given integers **a** and **b** (with **a ≠ 0**), we say **a divides b** if there exists an integer **c** such that:

- **b = ac**
- This means **b / a** is an integer.

- **Notation**:
  - **$a \mid b$ → a divides b**
  - **$a \nmid b$ → a does not divide b**

Key Terms

- **Factor (Divisor)**: If **$a \mid b$**, then **a** is a **factor** of **b**.
- **Multiple**: If **$a \mid b$**, then **b** is a **multiple** of **a**.

Mathematical Expression

- Using quantifiers:
  - **$\exists c \in \mathbb{Z}$ such that $ac = b$**

Examples

- **$3 \nmid 7$ → 7 ÷ 3 = 2.33** (Not an integer)
- **$3 \mid 12$ → 12 ÷ 3 = 4** (An integer)
- Let a, b, and c be integers, where a ≠ 0. Then :

  (i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

  (ii) if $a \mid b$, then $a \mid bc$ for all integers c;

  (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Division Cont..

For any integer **a** and positive integer **d**, there exist unique integers **q** (quotient) and **r** (remainder) such that:

$$a = dq + r, \quad 0 \le r < d$$

Key Terms & Notation

- **Dividend**: a, **Divisor**: d, **Quotient**: q, **Remainder**: r
- **Quotient**: $q=a \div d$, **Remainder**: $r=a \bmod d$

Modular Arithmetic

- Two integers **a** and **b** are **congruent modulo m** if **m divides a−b**:

$$a \equiv b \pmod{m} \text{ (means } m \mid (a-b) \text{)}$$

- **Notation**:

$$a \equiv b \pmod{m} \rightarrow \text{a and b are congruent modulo m}$$

$$a \not\equiv b \pmod{m} \rightarrow \text{a and b are not congruent modulo m}$$

- **Key Distinction**:

$$a \equiv b \pmod{m} \rightarrow \textbf{a relation}$$

$$a \bmod m = b \rightarrow \textbf{a function}$$

- Let a and b be integers, and let m be a positive integer. Then :

$$a \equiv b \pmod{m} \text{ if and only if } a \bmod m = b \bmod m.$$

Characterization and Operation on Congruence

- For a positive integer **m**, two integers **a** and **b** are congruent modulo **m** if and only if :

  a = b + km, for some integer k

  - This means **a and b differ by a multiple of m**.
  - The set of all integers congruent to **a mod m** is called its **congruence class**.
- If **a ≡ b (mod m)** and **c ≡ d (mod m)**, then :
  - Addition preserves congruence:

    $$a + c \equiv b + d \pmod{m}$$

  - **Multiplication preserves congruence**:

    $$ac \equiv bd \pmod{m}$$

Example :

Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows that :

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$ **and**

$$77 = 7 \cdot 11 \equiv 2 \cdot 1 = 2 \pmod{5}$$

Arithmetic modulo m

- Set **Z** :
    - Contains integers **{0, 1, ..., m - 1}**.
- **Addition Modulo m**:

$$a +_m b = (a + b) \mod m$$

- **Multiplication Modulo m**:

$$a \cdot_m b = (a \cdot b) \mod m$$

- These operations define **arithmetic modulo m**, which follows different rules than standard integer arithmetic.

Example :

Use the definition of addition and multiplication in Z to find

$$7 +_{11} 9 \text{ and } 7 \cdot_{11} 9.$$

Solution:

Using the definition of addition modulo 11, we find that

$$7 +_{11} 9 = (7 + 9) \mathbf{\ mod\ } 11 = 16 \mathbf{\ mod\ } 11 = 5,$$

and

$$7 \cdot_{11} 9 = (7 \cdot 9) \mathbf{\ mod\ } 11 = 63 \mathbf{\ mod\ } 11 = 8.$$

Hence, $7 +_{11} 9 = 5$ and $7 \cdot_{11} 9 = 8$.

Modular Exponentiation

- **Problem**: Computing b ^n mod m directly is inefficient due to large memory and time constraints.
- **Solution**: Use **binary exponentiation**, leveraging the binary representation of n.

- **Key Idea**: Instead of computing b^n first, break it down using:

$$b^n = b^{a_{k-1} \cdot 2^{k-1}} \cdots b^{a_1 \cdot 2^1} \cdot b^{a_0}$$

  - Compute powers efficiently:

$$b, \quad b^2, \quad (b^2)^2 = b^4, \quad (b^4)^2 = b^8, \quad \ldots$$

  - Multiply only the necessary terms where **binary digits of n are 1**.
  - **Reduce modulo m at each step** to save space and computation time.

Primes

- **Prime Number**: A number $p > 1$ is **prime** if its only positive factors are **1 and itself**.
- **Composite Number**: A number $n > 1$ that is **not prime** (i.e., it has divisors other than 1 and itself).

Fundamental Theorem of Arithmetic

- Every integer **n > 1** is **either prime or can be uniquely expressed** as a product of prime numbers.
- The prime factorization is unique, with factors listed in **nondecreasing order**.

Key Insights

- 1 is **not prime**, as it has only one factor.
- A number n is **composite** if there exists an integer a such that:

$1 < a < n$ **and** $a \mid n$.

There are infinitely many primes

GCD and LCM

Greatest Common Divisor (GCD) The greatest common divisor of two integers □ and □ , not both zero, is the largest integer □ such that:

□ | □ and □ | □

Example: gcd (24, 36)

- **The divisors of 24**: 1 , 2 , 3 , 4 , 6 , 8 , 12 , 24 1,2,3,4,6,8,12,24.
- **The divisors of 36:** 1 , 2 , 3 , 4 , 6 , 9 , 12 , 18 , 36 1,2,3,4,6,9,12,18,36.
- **Common divisors:** 1 , 2 , 3 , 4 , 6 , 12 1,2,3,4,6,12.

Largest common divisor: gcd (24 , 36) = 12

- The integers a and b are relatively prime if their greatest common divisor is 1.

LCM

The **least common multiple (lcm)** of two positive integers a and b is the smallest positive integer that is divisible by both a and b. It is denoted as:

lcm(a,b)

Existence of LCM:

- The set of numbers divisible by both a and b is nonempty (since ab is one such number).
- By the **well-ordering property** (every nonempty set of positive integers has a least element), the **smallest** such number must exist

LCM Using Prime Factorization:

If the prime factorization of a and b is:

$$a = p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n},$$

$$b = p_1^{b_1} p_2^{b_2} \ldots p_n^{b_n},$$

then the **least common multiple** is given by:

$$\operatorname{lcm}(a, b) = p_1^{\max(a_1, b_1)} p_2^{\max(a_2, b_2)} \ldots p_n^{\max(a_n, b_n)}.$$

LCM Cont..

This formula works because:

- A common multiple of aaa and bbb must contain at least the **maximum** number of each prime factor appearing in aaa and bbb.
- The **least** such multiple has no extra prime factors beyond those in a and b
- Alternative Formula Using GCD:

$$\mathrm{lcm}(a,b) = \frac{|a \cdot b|}{\gcd(a,b)}.$$

Linear Congruences

A linear congruence is of the form:

$$ax \equiv b \pmod{m}$$

where a, b, m are integers, and x is the unknown.

Solution Steps:

1. **Check if a solution exists**
   - A solution exists **if and only if** $\gcd(a, m) \mid b$
2. **Find the Modular Inverse (if $\gcd(a,m) = 1$)**
   - Find a^{-1} such that $aa^{-1} \equiv 1 \pmod{m}$
   - Use the **Extended Euclidean Algorithm** to compute a^{-1}
3. **Compute x**
   - Multiply both sides by a^{-1}

$$x \equiv a^{-1} b \pmod{m}$$

Example:

Solve 7 ▯ $\equiv 1 \pmod{26}$

- **gcd** $(7, 26) = 1$, so an inverse exists.
- $7^{-1} \equiv 15 \pmod{26}$
- ▯ $\equiv 15 \pmod{26}$

Pseudoprimes

FERMAT’S LITTLE THEOREM:

- If p is prime and a is an integer not divisible by p, then:

$$a^{p-1} \equiv 1 \pmod{p}.$$

- Furthermore, for every integer a we have $a^p \equiv a \pmod{p}$.

A **pseudoprime** is a composite number that satisfies a primality test for a specific base, even though it is not actually prime.

- **Base-2 Pseudoprimes:**

A composite number n is a pseudoprime to base 2 if:

$$2^{n-1} \equiv 1 \pmod{n}$$

even though n is not prime.

Example:

The smallest base-2 pseudoprime is 341:

$$2^{340} \equiv 1 \pmod{341}$$

but 341 = 11 × 31, so it is not prime.

Applications of Congruences

- **Hashing for Efficient Data Retrieval**
  A **hashing function** quickly assigns memory locations to records using a key k.
- **Pseudorandom numbers**

  Used in **computer simulations, cryptography, and gaming** when random-like behavior is needed.

- **Check Digits**

  Used for **error detection** in digit strings like **credit card numbers, barcodes, and ISBNs**.

Cryptography

The practice of securing information by transforming it so that only authorized parties can access it.

- Types of cryptography :

a.) Classical Cryptography: Caesar Cipher

A simple substitution cipher used by **Julius Caesar** to encrypt messages by shifting each letter **three places forward** in the alphabet.

CRYPTANALYSIS : The process of recovering plaintext from ciphertext without knowledge of both the encryption method and the key is known as cryptanalysis or breaking codes

b.) Public Key Cryptography

Is a cryptographic system that uses a pair of keys: a **public key** for encryption and a **private key** for decryption. Unlike private key cryptosystems, the encryption key does not reveal the decryption key, eliminating the need for secure key exchange.

RSA Cryptosystem

- In the RSA cryptosystem, each individual has an encryption key (n, e), where n = pq,
- Unfortunately, no one calls this the Cocks cryptosystem.
- modulus is the product of two large primes p and q, say with 300 digits each, and an exponent e that is relatively prime to (p – 1)(q – 1).
- To produce a usable key, two large primes must be found. This can be done quickly on a computer using probabilistic primality tests, referred to earlier in this section.
- However, the product of these primes n = pq, with approximately 600 digits, cannot, as far as is currently known, be factored in a reasonable length of time.
- As we will see, this is an important reason why decryption cannot, as far as is currently known, be done quickly without a separate decryption key.

RSA Encryption and Decryption

- To encrypt messages using a particular key (n, e), we first translate a plaintext message M into sequences of integers.
- To do this, we first translate each plaintext letter into a two-digit number, using the same translation we employed for shift ciphers, with one key difference.
- That is, we include an initial zero for the letters A through J, so that A is translated into 00, B into 01,..., and J into 09.
- Then, we concatenate these two-digit numbers into strings of digits.
- Next, we divide this string into equally sized blocks of 2N digits, where 2N is the largest even number such that the number 2525...25 with 2N digits does not exceed n. (When necessary, we pad the plaintext message with dummy Xs to make the last block the same size as all other blocks.)
- After these steps, we have translated the plaintext message M into a sequence of integers m1, m2,..., mk for some integer k.
- Encryption proceeds by transforming each block mi to a ciphertext block ci.
- This is done using the function :

c = m^e mod n.

<u>Decryption:</u>

- The plaintext message can be quickly recovered from a ciphertext message when the decryption key d, an inverse of e modulo (p − 1)(q − 1), is known

Cryptographic Protocols

Key Exchange

- Two parties can use to exchange a secret key over an insecure communications channel without having shared any information in the past.
- Generating a key that two parties can share is important for many applications of cryptography.
- For example, for two people to send secure messages to each other using a private key cryptosystem they need to share a common key.
- The protocol we will describe is known as the **Diffie-Hellman key agreement protocol**

Digital Signatures

- Not only can cryptography be used to secure the confidentiality of a message, but it also can be used so that the recipient of the message knows that it came from the person they think it came from.
- We first show how a message can be sent so that a recipient of the message will be sure that the message came from the purported sender of the message.
- In particular, we can show how this can be accomplished using the RSA cryptosystem to apply a **digital signature** to a message.